International Islamic University Chittagong

Department of Computer Science and Engineering

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Project Title

Bank Network Design & Implementation

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1. Introduction

A **bank network design project** aims to build a reliable, secure, scalable computer network for a bank's headquarters and branch offices. The network must support daily business operations, protect sensitive financial data, and allow secure communication between internal users, external systems, and the internet. This includes using modern networking concepts such as VLANs, routers, dynamic routing, and security mechanisms to provide robust connectivity and performance.

2. Background

Bank networks connect a variety of devices — computers, servers, ATMs, printers, and security systems — across departments like IT, Finance, HR, and Customer Service. They must be fault-tolerant, secure, and capable of handling high volumes of traffic with minimal downtime. VLANs are used to logically separate departments, and routing protocols like OSPF help traffic reach its destination efficiently.

3. Literature Review

Past works and examples show typical bank network designs using:

- **VLAN segmentation** to isolate departments and reduce broadcast domains for better performance and security.
- **Dynamic routing (OSPF)** to allow automatic sharing of routing information as network size grows.
- **SSH and security policies** to protect remote access and prevent unauthorized device access.
- **Redundancy and failover mechanisms** to ensure high availability.

These elements help networks scale and maintain reliability under increasing demand.

4. Problem Statement

Banks face challenges in designing networks that are:

- **Secure** enough to protect sensitive customer data.

- **Efficient** in data routing and resource usage.

- **Scalable** to support growing branches and services.
- **Reliable** with minimal downtime, even during failures.

The project must address these issues by using proper network design methodologies, segmentation, and routing techniques.

5. Designs

5.1 Topology

The network uses a hierarchical design:

- **Core Layer**: High-speed routers and switches connecting all departments.
- **Distribution Layer**: Layer 3 switches handling routing between VLANs.
- **Access Layer**: Switches and wireless access points connecting end devices.

5.2 VLAN Plan

Departments are separated into VLANs for security and performance:

VLAN 10: Management, VLAN 20: Research,VLAN 30: HR, VLAN 40:Marketing

VLAN 50:Accounting, VLAN 60:Finance, VLAN 70:Logistics, VLAN 80:Customer

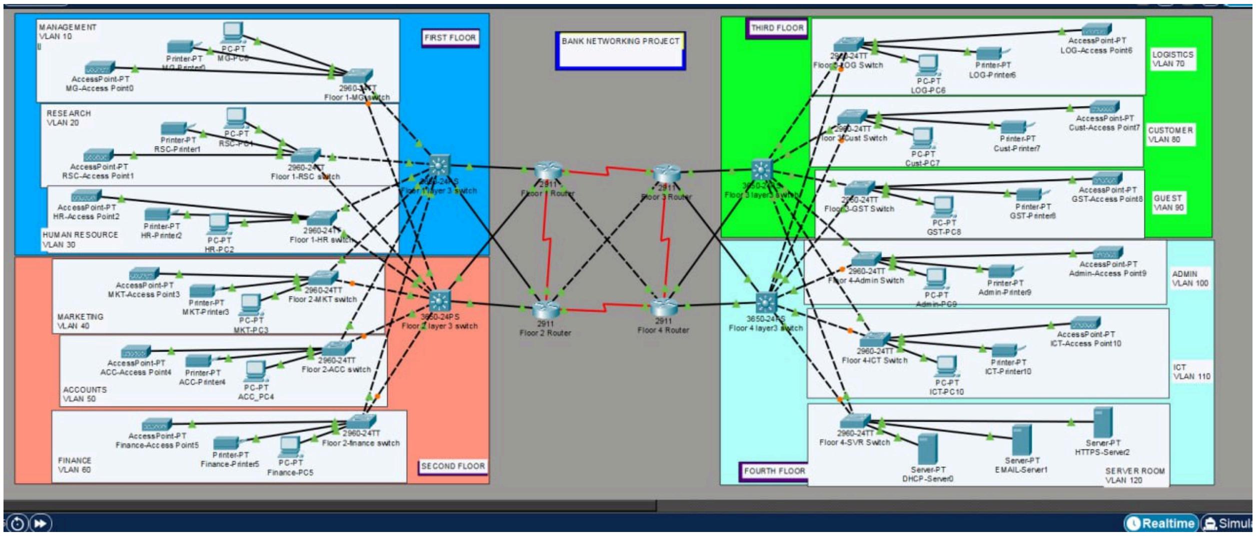
VLAN 90:Guest, VLAN 100:Admin, VLAN 110:ICT

5.3 IP Addressing Scheme

Hosts in each VLAN receive IPs from a defined subnet, either via DHCP or static assignment depending on role and security needs.

5.4 Routing

OSPF is deployed so routers and L3 switches automatically share routing information across VLANs and links.



6. Implementation

6.1 VLAN Configuration

Switches are configured with VLANs and assigned to specific ports based on department. Devices in the same VLAN can communicate directly, while inter-VLAN traffic goes through a L3 switch or router.

6.2 Routing and DHCP

- Configure OSPF on routers.
- Enable DHCP for dynamic IP assignment.
- Setup inter-VLAN routing using a Layer 3 switch or router-on-a-stick.

6.3 Security Features

Use port security, SSH logins, and access policies to protect devices and control access.

7. Experimental and Theoretical Results

7.1 Connectivity Testing

- **Ping tests** confirm hosts in different VLANs can communicate.
- **DHCP tests** show successful automatic IP allocation across VLANs.

7.2 Routing Verification

OSPF neighbors are established and routing tables show all VLAN subnets reachable.

7.3 Security Verification

Unauthorized access attempts are blocked by port security and SSH configurations.

8. Future Work

Future improvements may include:

- Better security with **firewalls and IPS/IDS**.
- SD-WAN integration for optimized branch connectivity.
- Implementing **network monitoring tools** for performance analysis.

9. Conclusions

The bank network design provides a secure, scalable, and efficient infrastructure using VLANs, dynamic routing (OSPF), DHCP, and proper security practices. Testing confirmed the design meets connectivity, segmentation, and accessibility requirements. The network is ready for deployment and future expansion.

10. References

1. Bank Network Design & Implementation, Packet Tracer project example.
2. Github example of bank network design with VLAN, OSPF, SSH.
3. VLAN segmentation purpose and use in networks.
4. SD-WAN banking network scalable solution.

5. Bank network project report example (IPv6, ACLs).

11. Appendix

A.Router Configuration

OSPF Configuration Example:

- OSPF process ID is set to 1
- The network 192.168.10.0 is advertised in area 0

VLAN Configuration Example:

- VLAN ID 10 is created
- The VLAN is assigned the name “Management”

B. IP Subnet Table

VLAN	Department	Subnet
10	Management	192.168.10.0/26
20	Customer	192.168.10.64/26
30	Finance	192.168.10.128/26